Q1. JEE Main 2026 (21 January Shift 2)

Let z be the complex number satisfying $|z-5| \leq 3$ and having maximum positive principal argument. Then $34\left|\frac{5z-12}{5iz+16}\right|^2$ is equal to :

(1) 26 (2) 12 (3) 20 (4) 16

Q2. JEE Main 2026 (23 January Shift 1)

Let $S=\{z: 3 \leqslant |2z-3(1+i)| \leqslant 7\}$ be a set of complex numbers. Then $\min_{z \in S}\left|\left(z+\frac{1}{2}(5+3i)\right)\right|$ is equal to :

(1) 2 (2) $\frac{5}{2}$ (3) $\frac{3}{2}$ (4) $\frac{1}{2}$

Q3. JEE Main 2026 (28 January Shift 2)

Let $A=\{z \in \mathbb{C}: |z-2| \leqslant 4\}$ and $B=\{z \in \mathbb{C}: |z-2|+|z+2|=5\}$. Then the max $\{|z_1-z_2|: z_1 \in$ A and $z_2 \in$ B$\}$ is :

(1) $\frac{17}{2}$ (2) 8 (3) 9 (4) $\frac{15}{2}$

Q4. JEE Main 2026 (24 January Shift 2)

Let $z=(1+i)(1+2i)(1+3i)\ldots(1+ni)$, where $i=\sqrt{-1}$. If $|z|^2=44200$, then n is equal to ____

Q5. JEE Main 2026 (22 January Shift 2)

Let $S=\left\{z \in \mathbb{C}: 4z^2+\bar{z}=0\right\}$. Then $\sum_{z \in S}|z|^2$ is equal to:

(1) $\frac{5}{64}$ (2) $\frac{1}{16}$ (3) $\frac{3}{16}$ (4) $\frac{7}{64}$

Q6. JEE Main 2026 (22 January Shift 1)

Let $\alpha=\frac{-1+i\sqrt{3}}{2}$ and $\beta=\frac{-1-i\sqrt{3}}{2}, i=\sqrt{-1}$. If
$(7-7\alpha+9\beta)^{20}+(9+7\alpha-7\beta)^{20}+(-7+9\alpha+7\beta)^{20}+(14+7\alpha+7\beta)^{20}=m^{10}$, then m is _____

Q7. JEE Main 2026 (23 January Shift 2)

If $z=\frac{\sqrt{3}}{2}+\frac{i}{2}, i=\sqrt{-1}$, then $\left(z^{201}-i\right)^8$ is equal to

(1) 1 (2) 256 (3) -1 (4) 0

Q8. JEE Main 2026 (21 January Shift 1)

If $x^2+x+1=0$, then the value of $\left(x+\frac{1}{x}\right)^4+\left(x^2+\frac{1}{x^2}\right)^4+\left(x^3+\frac{1}{x^3}\right)^4+\ldots+\left(x^{25}+\frac{1}{x^{25}}\right)^4$ is:

(1) 162 (2) 175 (3) 145 (4) 128

Q9. JEE Main 2026 (28 January Shift 1)

Let z be a complex number such that $|z-6|=5$ and $|z+2-6i|=5$. Then the value of $z^3+3z^2-15z+141$ is equal to

(1) 50 (2) 61 (3) 37 (4) 42

ANSWERS AND SOLUTIONS

1. (3) **2.** (3) **3.** (1) **4.** 5 **5.** (3) **6.** 49 **7.** (2) **8.** (3)
9. (1)

1. **(3)** The region $|z-5| \leq 3$ is a disk centered at $(5,0)$ with radius 3. Maximum positive principal argument from origin occurs at the tangent point from origin to the circle.

$\sin\theta = \frac{3}{5}, \cos\theta = \frac{4}{5}$.

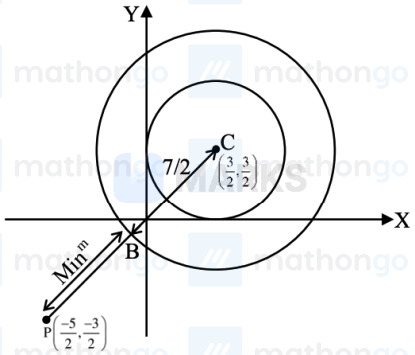
The tangent point is the foot of perpendicular from $(5,0)$ to the line $3x-4y=0$, giving $z = \frac{16}{5} + \frac{12}{5}i$.

$5z - 12 = 4 + 12i \Rightarrow |5z-12|^2 = 160$

$5iz + 16 = 4 + 16i \Rightarrow |5iz+16|^2 = 272$

$34\left|\frac{5z-12}{5iz+16}\right|^2 = 34 \times \frac{160}{272} = 34 \times \frac{10}{17} = 20.$

2. **(3)** $\frac{3}{2} \leq \left|z - \frac{3}{2}(1+i)\right| \leq \frac{7}{2}$



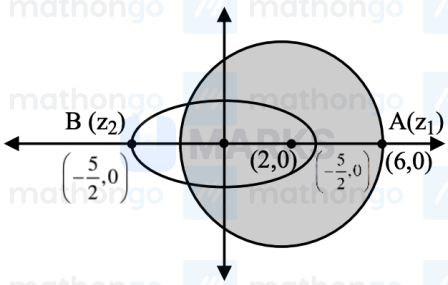
$\min_{z\in S}\left|z - \left(\frac{-5}{2} - \frac{3}{2}i\right)\right| = PB$

$PB = PC - \frac{7}{2} \Rightarrow 5 - \frac{7}{2} \Rightarrow \frac{3}{2}$

3. **(1)** $|z-2| \leq 4 \Rightarrow (x-2)^2 + y^2 \leq 16$

$|z-2| + |z+2| = 5 \Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\Rightarrow \frac{4x^2}{25} + \frac{4y^2}{9} = 1$



Maximum value of $|z_1 - z_2| = 6 + \frac{5}{2} = \frac{17}{2}$

4. **(5)** $z = (1+i)(1+2i)(1+3i)\cdots(1+ni)$ where $i = \sqrt{-1}$
$|z|^2 = \prod_{k=1}^{n} |1+ki|^2 = \prod_{k=1}^{n}(1+k^2)$
Computing: $(1+1)(1+4)(1+9)(1+16)(1+25) = 2\times 5\times 10\times 17\times 26$
$= 10\times 10\times 17\times 26 = 100\times 442 = 44200$
Therefore $n = 5$

5. **(3)** Let $z = x+iy$. From $4z^2 + \bar{z} = 0$, separating real and imaginary parts:
Real: $4x^2 - 4y^2 + x = 0$, Imaginary: $y(8x-1) = 0$.
Case 1: $y = 0$
$\Rightarrow x(4x+1) = 0$
$\Rightarrow z = 0$ or $z = -\frac{1}{4}$.
Case 2: $x = \frac{1}{8}$
$\Rightarrow \frac{1}{16} - 4y^2 + \frac{1}{8} = 0$
$\Rightarrow y^2 = \frac{3}{64}$.
So $|z|^2 = \frac{1}{64} + \frac{3}{64} = \frac{1}{16}$.
$\sum |z|^2 = 0 + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} = \frac{3}{16}$

6. **(49)** $\alpha = \omega$, $\beta = \omega^2$ (cube roots of unity).
Let $z_1 = 7 - 7\omega + 9\omega^2$, $z_2 = 9 + 7\omega - 7\omega^2$, $z_3 = -7 + 9\omega + 7\omega^2$.
Calculating: $z_1 = 6 - 8i\sqrt{3}$, $z_2 = 9 + 7i\sqrt{3}$, $z_3 = -15 + i\sqrt{3}$.
All have $|z_k| = \sqrt{228}$.
Verify: $z_2 = z_1\omega$, $z_3 = z_1\omega^2$.
$z_1^{20} + z_2^{20} + z_3^{20} = z_1^{20}(1+\omega^{20}+\omega^{40}) = z_1^{20}(1+\omega^2+\omega) = 0$.
Fourth term: $14 + 7\omega + 7\omega^2 = 14 + 7(-1) = 7$. So $7^{20} = 49^{10}$.
Total $= 0 + 49^{10} = m^{10} \Rightarrow m = 49$.

7. **(2)** $z = \frac{\sqrt{3}}{2} + \frac{i}{2} = e^{i\pi/6}$.
$z^{201} = e^{i\cdot 201\pi/6} = e^{i\cdot 33.5\pi} = e^{i(33\pi+\pi/2)}$.
Since $e^{i\cdot 33\pi} = e^{i\pi} = -1$, we get $z^{201} = -1\cdot e^{i\pi/2} = -i$.
$z^{201} - i = -i - i = -2i$.
$(z^{201} - i)^8 = (-2i)^8 = 2^8\cdot i^8 = 256\cdot 1 = 256$.

8. **(3)** Given $x^2 + x + 1 = 0$, so $x = \omega$ (primitive cube root of unity) with $x^3 = 1$.
From the equation: $x + \frac{1}{x} = -1$.
Since $x^3 = 1$, value of $x^n + x^{-n}$ has period 3:
$n \equiv 0 \pmod 3$: $x^n + x^{-n} = 2$
$n \equiv 1, 2 \pmod 3$: $x^n + x^{-n} = -1$
Fourth powers: $(2)^4 = 16$, $(-1)^4 = 1$.
For $n = 1$ to 25: 8 multiples of 3 contribute $8\times 16 = 128$, remaining 17 terms contribute $17\times 1 = 17$.
Total $= 128 + 17 = 145$.

9. **(1)** From $|z-6|=5$, point z lies on circle centered at $(6,0)$ with radius 5.
From $|z+2-6i|=5$, point lies on circle centered at $(-2,6)$ with radius 5.
Let $z=x+iy$.
First condition gives: $x^2+y^2=12x-11$.
Second gives: $x^2+y^2=-4x+12y-15$.
Setting equal: $16x-12y+4=0$, so $x=\frac{3y-1}{4}$.
Substituting into $(x-6)^2+y^2=25$: $(3y-25)^2+16y^2=400$, which simplifies to $25y^2-150y+225=0$, giving $y=3$ and $x=2$.
Thus $z=2+3i$, so $z^2=-5+12i$ and $z^3=-46+9i$.
$z^3+3z^2-15z+141=(-46+9i)+3(-5+12i)-15(2+3i)+141=50$